STEP 8

Vibe Coding and Automation

Fireflies.ai Product Management Launchpad Capstone

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Purpose: use a lightweight interactive prototype to make the proposed commitment lifecycle tangible, test key interactions, and expose product assumptions before engineering investment.

1. Executive Summary

Step 8 uses vibe coding as a product-learning tool, not as proof that the proposed solution works in production. The prototype is intended to demonstrate how a commitment can be captured, assigned with consent, updated, reassigned, and resolved honestly. It should help reviewers assess whether the experience is understandable and lightweight before the team invests in production integrations or broader automation.

The prototype work described in this report includes a Fireflies-style commitments interface and a manual task-creation flow. The reported implementation includes fields for title, description, supporting links or attachment metadata, assignee, due date, priority, related meeting, completion criteria, and notes. These are prototype-level capabilities; they do not establish that uploaded files are stored durably, document contents are parsed, or tasks synchronize with external systems.

What the prototype can help evaluate	What it cannot establish by itself
Whether key actions and status meanings are understandable.	Whether users will adopt the workflow in real teams.
Whether owner confirmation and status changes are easy to demonstrate.	Whether the experience raises the 14-day resolution rate.
Whether the task form exposes missing information and excessive friction.	Whether backend persistence, permissions, notifications, and integrations are production-ready.
Whether the UI behaves reasonably across representative screen sizes when tested.	Whether performance, accessibility, security, and reliability meet production requirements.

2. Objective and Decisions

The objective is to build the smallest credible prototype that tests the riskiest interaction assumptions from Steps 3–7.

- Can a person tell when a commitment is only suggested versus when ownership is confirmed?
- Can users distinguish Done, Deliberately dropped, Reassignment pending, Accepted reassignment, and Blocked/Waiting?
- Can a commitment be reassigned without counting it as resolved before the new owner accepts?
- Can a user manually capture a commitment that was missed by extraction without turning Fireflies into a full project-management tool?
- Can the experience work on smaller screens without hiding essential actions or creating horizontal overflow?

Decision boundary: this step validates interaction concepts and prototype usability. It does not reopen the core product direction or claim that automation has been proven to improve outcomes.

3. Prototype Scope

Area	Prototype intent	Acceptance signal for review
Commitment list	Show commitments and their current states in a scannable interface.	A reviewer can identify what needs attention and what state each item is in.
Owner confirmation	Distinguish suggested ownership from confirmed ownership; provide a path to accept, decline, or clarify.	No task appears confirmed solely because a person was suggested.
Status controls	Represent Done, Deliberately dropped, Blocked/Waiting, and reassignment states distinctly.	A blocked item or pending handoff is not visually or analytically treated as resolved.
Reassignment	Show a transfer request and separate acceptance by the new owner.	Reassignment qualifies as resolved only after acceptance.
Manual task creation	Allow a user to enter task details and relevant meeting/context references.	Required fields validate; task appears in the prototype list and details view.
Responsive layout	Keep key task content and actions usable at desktop and mobile widths.	Representative viewport checks show no essential action hidden and no unintended horizontal overflow.

4. Manual Task Creation: Supporting Capability

Manual task creation supports the main commitment lifecycle by giving users a way to capture an action when it is not extracted or when it needs to be recorded manually. It is a supporting capability, not a third core product bet.

Field	Product guidance
Title	Required; use an action-oriented description where possible.
Description and notes	Optional context; avoid forcing users to repeat information already available from the meeting.
Assignee	Allow a suggested or selected owner, but make confirmation status explicit; do not imply acceptance.
Due date and priority	Optional unless the source commitment or workflow establishes them; do not invent deadlines.
Related meeting	Link the task to its source meeting when known.
Completion criteria	Capture only when useful to determine what Done means; allow clarification.
Links and attachment metadata	Support references in the prototype. Do not represent metadata or a link as proof of file upload, durable storage, parsing, or permission-safe access.

5. Automation Principles

Automation should reduce repetitive effort without hiding uncertainty or taking away user control.

- Prefill only information supported by the source context; mark uncertain owner or deadline fields as suggestions.
- Ask the user to confirm, decline, or clarify a proposed owner instead of silently assigning responsibility.
- Keep Blocked/Waiting non-terminal and make it possible to record a dependency without blaming the owner.
- Use reminders sparingly and contextually; do not assume that more notifications will improve closure.
- Maintain an understandable activity history for ownership and status changes.
- Treat external integrations, automatic document parsing, and production notification delivery as future dependencies unless implemented and verified.

6. Validation Plan

A prototype walkthrough should test observable behavior with representative participants. Do not substitute the builder's own successful click-through for user evidence.

Test task	What to observe	Failure signal
Review a newly extracted commitment.	Can the participant tell whether the owner is suggested or confirmed?	Participant assumes the suggested owner has already accepted.
Accept, decline, or clarify ownership.	Can the participant find and explain the available choices?	Participant cannot decline without appearing to fail or feels forced to accept.
Mark work blocked by a dependency.	Does the participant understand that Blocked/Waiting is not closure?	Blocked is mistaken for Done or treated as owner failure.
Reassign a commitment.	Does the participant understand that the handoff remains pending until accepted?	The task is treated as resolved immediately when the request is sent.
Mark a commitment deliberately dropped.	Does the participant understand this as an intentional outcome distinct from Done?	Drop is hidden, stigmatized, or used as a generic way to clear overdue work.
Create a manual task with context.	Can the participant complete the form without unnecessary fields or confusion?	The form feels like duplicate project-management data entry.
Use the experience on a mobile viewport.	Can the participant find primary actions and read task content?	Important actions are hidden, crowded, or require horizontal scrolling.

7. Technical and Product Limitations

- Prototype state is not evidence of production persistence. Confirm where task data is stored, whether it survives refresh or a new session, and how multi-user updates would work.
- Links and attachment metadata do not prove that file bytes are uploaded, stored, parsed, or available to all authorized users.

- A local prototype does not prove real Fireflies authentication, permissions, event delivery, notification scheduling, or integrations with Jira, CRM, Smartsheet, or other systems.
- Type-checking, build success, and manual click-throughs—when performed—are useful implementation checks, but they are not usability findings or evidence of business impact.
- Any usability test result must be reported with participant count, role, tasks attempted, observed failures, and limitations. No such results should be claimed until the tests have actually been conducted and documented.

8. Definition of Prototype Readiness

- Core lifecycle states and transitions match the product definition used in Steps 3–7.
- Suggested ownership is visually distinct from accepted ownership.
- Reassignment is pending until accepted by the new owner.
- Done, Deliberately dropped, Blocked/Waiting, and pending handoff are not conflated.
- Manual task creation validates required fields and preserves entered context within the prototype's actual capabilities.
- Primary interactions work at desktop and mobile viewport sizes selected for testing.
- Known limitations are documented and no prototype-only capability is presented as production-ready.
- A small usability walkthrough has been run—or is explicitly scheduled—before the design is treated as validated.

9. Decisions and Next Actions

Decision / action	Why it matters	Evidence needed
Keep manual task creation supporting, not core.	Protects focus on ownership and honest closure.	Observe whether users need manual capture and whether it adds friction.
Test owner confirmation and resolution semantics first.	These are the riskiest assumptions in the product direction.	Task-based usability observations and participant explanations.
Keep attachments and integrations as explicit prototype limitations.	Prevents a visual demo from being mistaken for a production capability.	Technical verification of storage, access, and integration behavior before making claims.
Do not claim resolution-rate impact from prototype completion.	UI functionality does not establish behavioral or business outcomes.	Instrumented pilot with mature 14-day cohorts and a credible comparison.
Carry the validated flow into the Step 9 story.	Ensures the final presentation reflects the actual product concept and known limitations.	Prototype walkthrough, observed findings, and an honest limitations slide.

10. Recommendation

Use the prototype to test the commitment lifecycle through task-based user testing. Prioritize clear ownership confirmation, honest resolution states, and low-friction interactions. Document usability findings and technical limitations before including the prototype in the final presentation. Do not claim business impact without measured results.

11. Prototype Audit: Verification Evidence and Limitations

Audit conclusion. The commitment-management prototype is suitable for a capstone demonstration after a small startup/configuration fix, with the limitations in this section disclosed. This is not a production-readiness approval, a usability validation, or evidence of business impact.

11.1 Reproduced issue and repair

The previously reported “SyntaxError: Illegal return statement” was not reproduced in the current checkout. The reproducible failure occurred before application code loaded: Vite could not resolve/load the app's vite.config.ts and reported an access-denied directory read in the current Windows environment. The package.json development, build, and preview scripts were updated to use --configLoader runner. No UI, route, data-model, or business-logic changes were made.

11.2 Verification results

Area	Result	Evidence / boundary
Commitment app typecheck	PASS	TypeScript completed successfully.
Production build	PASS	Vite transformed 1,684 modules.
Development startup	PASS	Vite started at localhost:5173.
Browser console/page errors	PASS	No console or page errors after reload.
Task creation and title validation	PASS	Task appeared under Manual tasks; empty title showed validation.
Ownership confirmation	PASS	Candidate became Open and candidate count decreased.
Resolution states	PASS	Done and Deliberately dropped appeared as distinct resolved outcomes.
Blocked state	PASS	Dependency required; task remained unresolved.
Reassignment	PASS — simulated	Pending acceptance was shown; acceptance closed original as Reassigned and opened linked instance.
Filters and counts	PASS with terminology caveat	Counts updated correctly; labels differ from preferred terminology.
Responsive layout	PASS within tested scope	No horizontal overflow at 1440px, 768px, or 390px.
Full workspace typecheck/build	FAIL	The separate fireflies-commitments-deck package is missing. sdm modules/type errors.

11.3 Product-semantic checks

- Resolved states: Done, Deliberately Dropped, and Reassigned.
- Unresolved states: Candidate, Open, Blocked, and Awaiting Acceptance.
- A reassignment remains pending until acceptance is simulated; the original then becomes Reassigned and the linked instance becomes Open.
- The current UI labels are “Candidates,” “Open / blocked / transfer,” and “Resolved outcomes.” These differ from “Needs Confirmation,” “Open,” and “Resolved.” This is a terminology discrepancy; the audit did not identify a state-counting defect.

11.4 Limitations and what the audit does not prove

- Reassignment acceptance is simulated in one browser; it does not verify a real second-user handoff.
- There is no backend, authentication, real multi-user identity, live synchronization, notifications, or external integration in this prototype.

- The full workspace build remains blocked by unrelated errors in the separate slide-deck package, despite the commitment app's typecheck and production build passing.
- Git status could not be verified because the .git directory was not a valid repository.
- Mobile viewport checks covered overflow and primary-action visibility at 390px; mobile-specific dialog behavior was not separately tested.
- The audit is a technical and interaction check, not a usability study. No participant usability results or business impact should be inferred from it.

11.5 Recommended next actions

- Normalize filter labels to the product vocabulary: Needs Confirmation, Open, and Resolved.
- Run task-based usability sessions with documented participant roles, tasks, observed errors, and outcomes before calling the experience user-validated.
- Test reassignment with distinct authenticated users and backend-persisted state before any production-readiness claim.
- Add automated tests for the terminal resolution states, pending reassignment, and the 14-day resolution event definition.
- Keep product impact as a hypothesis until a controlled pilot measures the 14-day commitment resolution rate and guardrails.

11.6 Presentation-safe summary

Use this wording: “The commitment app typecheck and production build passed after a Vite config-loader workaround. Core lifecycle interactions and responsive overflow were checked. Reassignment is simulated, the full workspace still has unrelated slide-deck build errors, and no usability or business-impact claim is made.”